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Using the scopes of cyber protection, what would you have done to prevent StorageCraft's backup data loss?

1. Safety:

To ensure safety, I would have implemented a staged migration process where backup data is duplicated and verified before transferring operations to a new platform. This means maintaining a temporary failover backup in both the internal data center and Google Cloud until the new system is confirmed stable. Additionally, I would require a full data integrity check after migration to confirm that no client backup data has been lost. This prevents accidental data loss caused by human error during system transitions.

2. Accessibility:

I would ensure that backup data remains consistently accessible to clients by maintaining redundant storage locations and not shutting down the original data center until the cloud backup is fully validated. Proper documentation and monitoring tools should be used to track backup availability in real time. By performing gradual migration rather than moving everything at once, the service remains operational and clients retain access to their backup data even during system changes.

3. Privacy:

While the incident did not affect primary data, protecting backup data also protects the privacy of client information. I would enforce encryption both in transit and at rest to ensure that even during migration, no unauthorized individuals can access the data. Regular privacy audits should also be performed to verify that data handling and storage procedures meet compliance standards. This way, privacy remains protected even if an error occurs.

4. Authenticity:

To maintain authenticity, I would require a checksum or hash verification process before and after data migration to confirm that all backup data remains unchanged. This ensures that backup files are genuine and have not been corrupted or altered during transfer. I would also maintain detailed logs of data changes and access so that integrity can be traced and verified if an issue occurs. These steps preserve data correctness and trustworthiness.

5. Security:

I would strengthen security by implementing strict access controls, ensuring that only authorized and trained personnel are allowed to manage backup migration processes. Additionally, I would use automated backup validation tools to detect and prevent accidental deletion or overwrite of stored data. Regular disaster recovery drills and security tests would help ensure that the backup system is resilient against both human error and cyber threats. This approach protects backup data from unintended loss.